The University of the West Indies

COMP6940: Big Data and Data Visualization

Group Assignment 2 Report

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Group Members

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1 Introduction

This assignment consists of two main tasks focused on applying advanced data analytic techniques to real-world problems. The first task involves modelling time-to-failure data using a Weibull distribution with both CPU and GPU implementations. The second task focuses on building and interpreting a credit risk classification model using machine learning techniques.

In **Part 1**, the objective is to estimate the parameters of a two-parameter Weibull distribution using maximum likelihood estimation on turbofan engine failure data from the NASA C-MAPSS dataset. Both a traditional CPU-based implementation and a GPU-accelerated implementation using JAX were used, allowing a performance comparison and demonstrating the value of accelerated computing in statistical modelling.

In **Part 2**, the objective was to build a credit default prediction model using the Home Credit Default Risk dataset, evaluate its financial impact, interpret its predictions using SHAP and counterfactual explanations, and assess fairness and responsible use considerations.

2 Part 1 — GPU-Accelerated Weibull Fitting

2.1 Parameter Estimation

The NASA C-MAPSS FD001 dataset was used to extract time-to-failure values for 100 turbofan engines. For each engine, the maximum cycle count was taken as the failure time, producing a dataset of observed lifetimes.

A two-parameter Weibull distribution was fitted to the data using maximum likelihood estimation. The log-likelihood function was implemented in both NumPy and JAX, with gradients computed using automatic differentiation in JAX.

The estimated parameters were:

- Shape parameter: $k = 4.408526$
- Scale parameter: $\lambda = 225.025671$

The value of $k > 1$ indicates an increasing failure rate over time, suggesting a wear-out failure mechanism. This aligns with expectations for mechanical systems such as turbofan engines, where components degrade through continued use. The scale parameter λ represents the characteristic life, meaning that approximately 63.2% of engines are expected to fail by this cycle count.

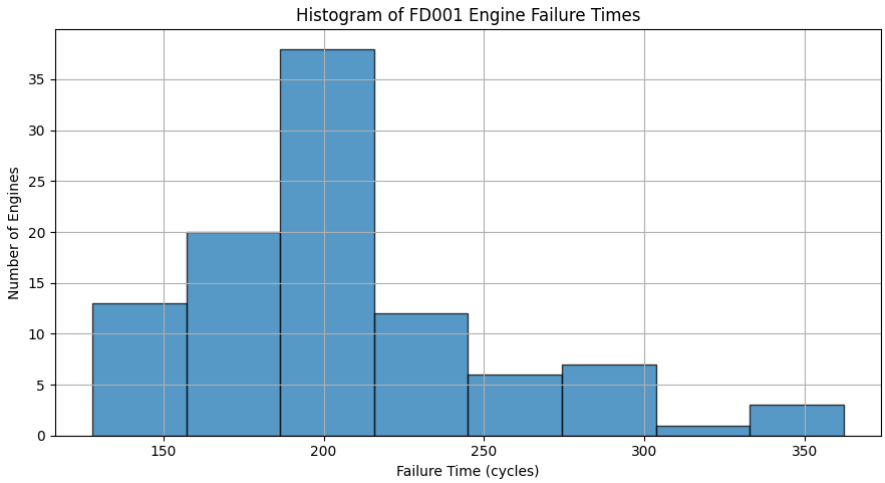

Figure 1. Histogram of FD001 engine failure times.

Figure 1 shows the distribution of the 100 observed engine failure times. Most engines failed between roughly 150 and 230 cycles, with fewer engines surviving to much higher cycle counts. This suggests that failures are concentrated in a central range, with a slight right tail caused by a smaller number of longer-lasting engines.

2.2 Fit Quality Assessment

The fitted Weibull distribution was evaluated by comparing the estimated probability density function to the histogram of observed failure times.

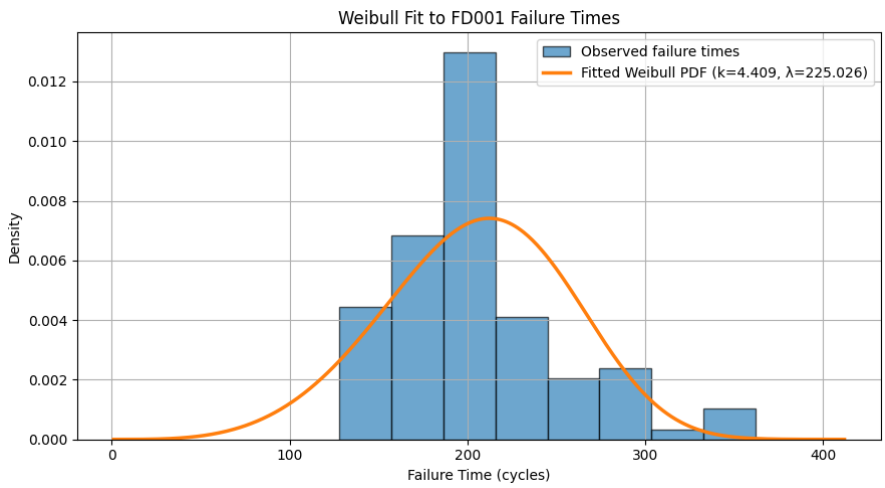

Figure 2. Observed failure times with fitted Weibull PDF overlay.

Figure 2 overlays the fitted Weibull probability density function on the observed failure-time histogram. The model provides a good fit in the central region of the data, where most observations are concentrated. The fitted curve follows the overall shape of the histogram reasonably well, which suggests that the Weibull distribution captures the main failure behaviour of the engines.

However, some limitations in the fit are also visible. The tails are not matched perfectly, meaning that very early or very late failures are not captured as well as the central mass of the data. This

is not unexpected, especially given that the sample contains only 100 engines and the Weibull model assumes a single underlying failure mechanism.

Overall, the Weibull distribution provides a reasonable and interpretable approximation of the failure data.

2.3 Maintenance Recommendation

Using the fitted Weibull survival function, a preventive maintenance threshold was determined based on the condition $R(t) = 0.9$, meaning that 90% of engines are expected to survive up to this cycle count.

From the model, this corresponds to approximately:

- $t \approx 135.07$ cycles

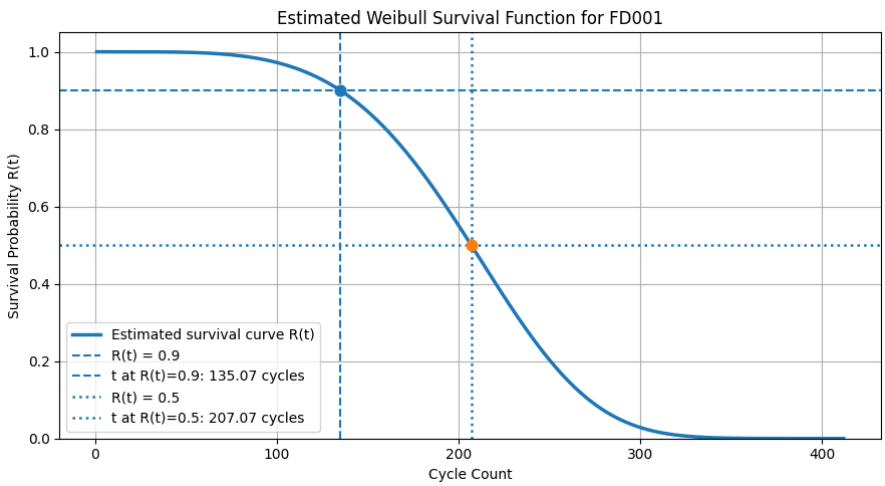

Figure 3. Estimated Weibull survival function with reliability thresholds.

Figure 3 shows the estimated survival function $R(t)$ for the fitted Weibull model. The point where $R(t) = 0.9$ occurs at approximately 135.07 cycles, which gives a practical preventive maintenance threshold if the goal is to keep at least 90% of engines operating without failure. The figure also shows the point where $R(t) = 0.5$, which occurs at approximately 207.07 cycles and represents the median life of the engines.

Scheduling maintenance before 135.07 cycles ensures a low probability of in-service failure, approximately 10%, which is critical in safety-sensitive systems such as aircraft engines. This recommendation reflects a conservative reliability strategy. However, in practice, the optimal maintenance schedule depends on the trade-off between maintenance costs and failure costs, so different operational contexts may justify different reliability thresholds.

2.4 GPU vs CPU Performance

The GPU-based implementation using JAX demonstrated improved computational efficiency compared to the CPU-based approach. By leveraging automatic differentiation and parallel computation, the GPU implementation was able to compute gradients and optimize the log-

likelihood more efficiently. This highlights the advantage of GPU acceleration for iterative optimization problems, particularly when scaling to larger datasets.

Although the dataset used in this analysis is relatively small, the performance gains are expected to become more significant for larger-scale applications.

3 Part 2 — Classification, Impact Simulation, and Model Interpretability

The second part of the assignment focused on predicting loan default using the Home Credit Default Risk dataset. The goal was not only to build a classifier with strong predictive performance, but also to evaluate whether the model was financially useful, interpretable, and fair. This makes the analysis more realistic, since a credit model should be judged by business value and responsible use as well as by statistical performance.

The dataset used was `application_train.csv`, where the target variable `TARGET` indicates whether an applicant defaulted on the loan. A value of 1 represents default and a value of 0 represents non-default. Because the dataset is imbalanced, multiple evaluation metrics were used rather than accuracy alone.

3.1 Data Preparation and Feature Engineering

The raw application data was cleaned according to the assignment requirements. Columns with more than 40% missing values were removed. Remaining numeric variables with missing values were imputed using the median, and remaining categorical variables with missing values were imputed using the mode. Rows with `AMT_INCOME_TOTAL` above the 99.5th percentile were removed as extreme outliers, and rows with `DAYS_EMPLOYED = 365243` were removed because this is a known anomaly in the dataset.

Five engineered features were created to better capture affordability and household context:

- `credit_income_ratio` = `AMT_CREDIT / AMT_INCOME_TOTAL`
- `annuity_income_ratio` = `AMT_ANNUITY / AMT_INCOME_TOTAL`
- `credit_term` = `AMT_CREDIT / AMT_ANNUITY`
- `days_employed_ratio` = `DAYS_EMPLOYED / DAYS_BIRTH`
- `income_per_family` = `AMT_INCOME_TOTAL / CNT_FAM_MEMBERS`

These engineered variables were useful because they express borrowing burden more directly than the original variables alone. Categorical variables were then one-hot encoded so that all models could be trained on a consistent numerical feature set.

3.2 Model Training and Evaluation

The curated dataset was split into training and test sets using an 80/20 stratified split so that the class distribution was preserved. Two models were trained:

- Logistic Regression
- XGBoost Classifier

This allowed comparison between a simple linear baseline and a more flexible gradient-boosted tree model. Performance on the held-out test set was evaluated using the required metrics:

- ROC-AUC
- Matthews Correlation Coefficient (MCC)
- Precision
- Recall
- F1-score

The XGBoost model was selected for the remaining analysis because it provided the strongest overall predictive performance and was better suited to capturing nonlinear relationships in the data. This was important because credit default behaviour is unlikely to be fully linear, and a more flexible model can better capture complex interactions between affordability, external credit indicators, and behavioural variables.

3.3 Business Impact Simulation

Predictive quality alone does not show whether a lending strategy is useful in practice, so the selected model was also evaluated through a business impact simulation on the held-out test set.

The assumptions used were:

- Profit from a correctly approved non-defaulter: 10% of loan amount
- Loss from an incorrectly approved defaulter: 50% of loan amount
- Opportunity cost from rejecting a good applicant: 10% of loan amount

Applicants were treated as approved when the predicted probability of default was below the classification threshold, and rejected otherwise. Expected profit was computed at thresholds of 0.3, 0.5, and 0.7.

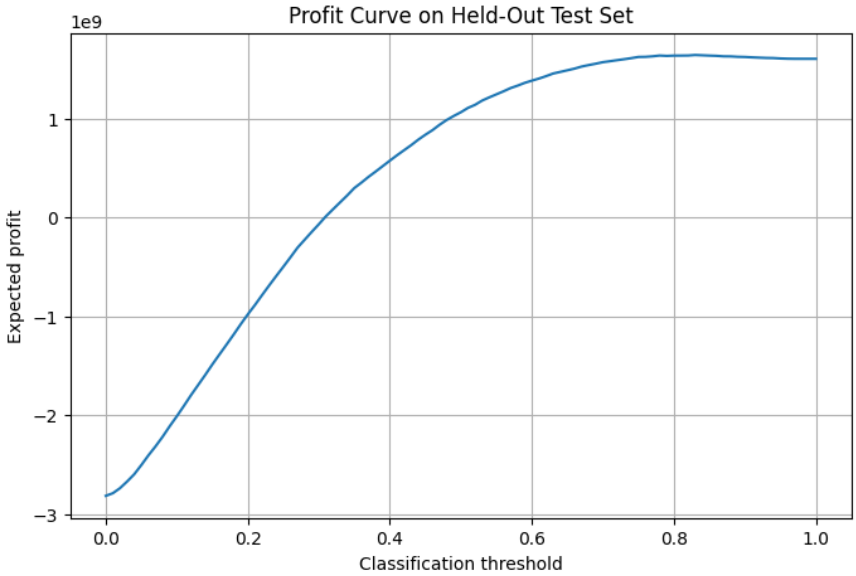

Figure 4. Profit curve on the held-out test set across classification thresholds.

The profit curve shows how expected profit changes as the threshold varies. At very low thresholds, too many applicants are rejected, including many good borrowers, so expected profit is strongly negative. As the threshold increases, profit rises sharply because more good borrowers are approved. Beyond the maximum region, the curve flattens because approving too many applicants also increases losses from default. This result shows clearly that the best decision threshold should be selected using economic value rather than classification metrics alone.

The selected model was also compared with two baselines:

- Approve everyone
- A random classifier with the same approval rate

This comparison is important because a useful machine learning model should outperform naive alternatives. In our analysis, the selected model improved on both simple strategies, which supports its practical use as a decision-support tool.

A sensitivity analysis was also carried out on the assumed bad-loan loss rate.

Figure 5. Sensitivity of expected profit to the assumed bad-loan loss rate.

The sensitivity plot shows that expected profit declines steadily as the assumed loss on incorrectly approved defaulters rises from 0.30 to 0.70. This confirms that the financial value of the model depends strongly on the institution's actual loss structure. The broader conclusion remains unchanged: threshold selection should be tied to realistic business costs rather than treated as a purely statistical tuning step.

3.4 SHAP Explanations

To interpret the selected model, SHAP was used for both global and local explanations. A random subset of 1,000 test observations was used for the global analysis.

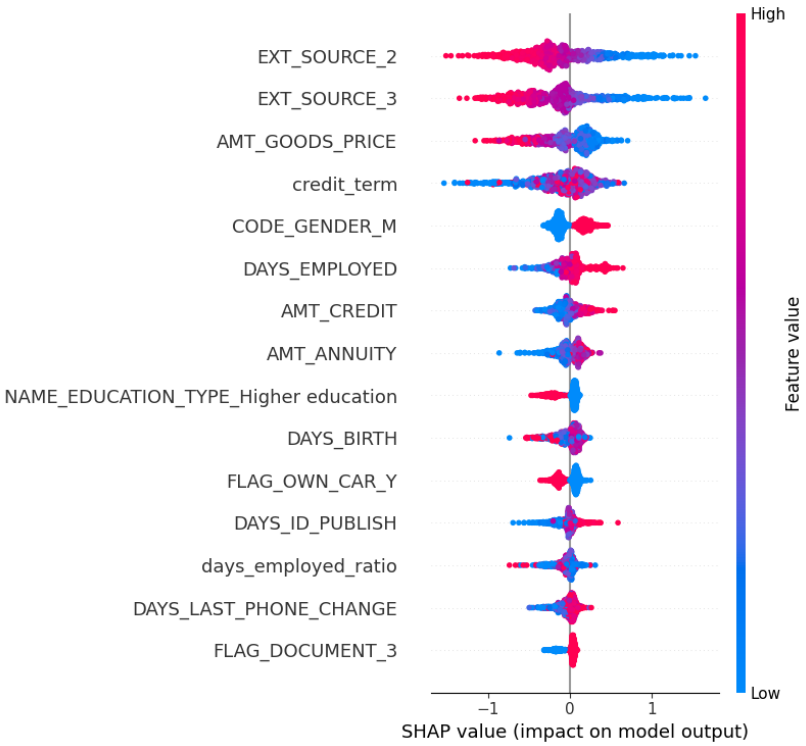

Figure 6. SHAP summary plot showing the top 15 features influencing the best model.

The SHAP summary plot shows that `EXT_SOURCE_2` and `EXT_SOURCE_3` were the most influential variables in the model. Lower values of these external risk scores tended to push predictions toward default, while higher values reduced predicted risk. Other important variables included `AMT_GOODS_PRICE`, `credit_term`, `DAYS_EMPLOYED`, `AMT_CREDIT`, `AMT_ANNUITY`, and `DAYS_BIRTH`. This indicates that the model combined external credit indicators with direct measures of loan burden and employment stability.

Two individual SHAP waterfall plots were then examined: one for a correctly classified defaulter and one for a correctly classified non-defaulter.

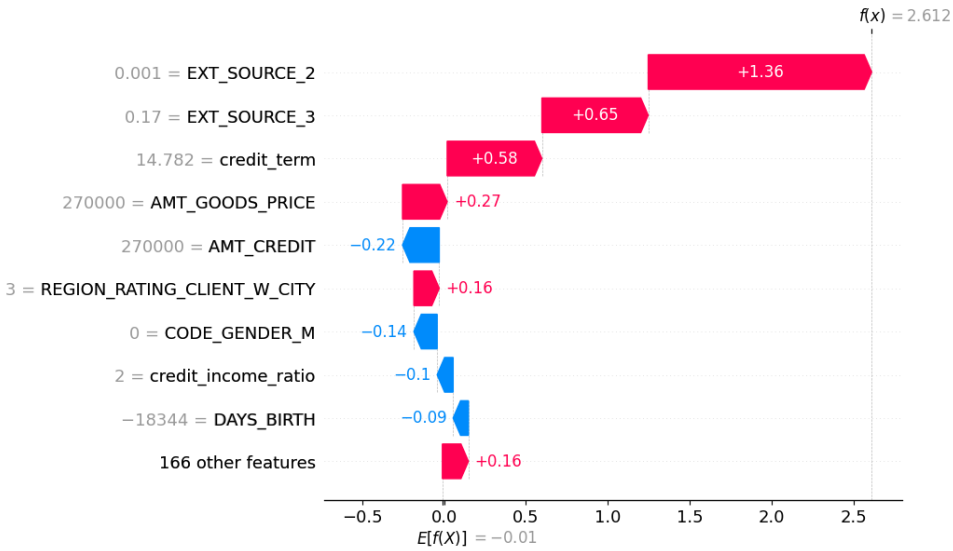

Figure 7. SHAP waterfall plot for a correctly classified defaulter.

The waterfall plot for the correctly classified defaulter shows that the largest contributions toward default came from very poor values of `EXT_SOURCE_2` and `EXT_SOURCE_3`, together with upward pressure from the loan profile. Features such as goods price and a less favourable credit term also increased the risk score, while some other features partially reduced it. This means the model classified the applicant as risky mainly because the external risk indicators were weak and the overall borrowing context was unfavourable.

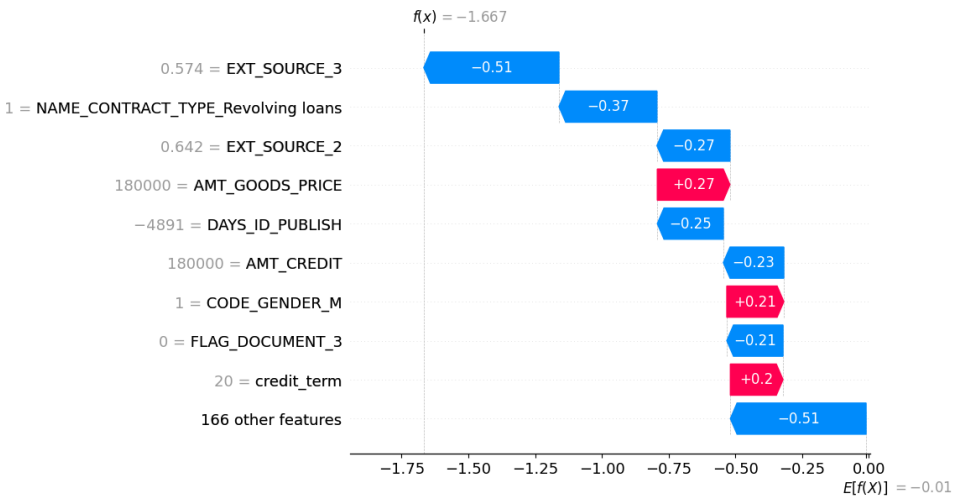

Figure 8. SHAP waterfall plot for a correctly classified non-defaulter.

The waterfall plot for the correctly classified non-defaulter shows that favourable external risk scores pushed the prediction strongly away from default. Positive contributions from `EXT_SOURCE_2` and `EXT_SOURCE_3` dominated the explanation, while other features such as the loan profile and age-related factors had smaller effects. This suggests that external scores, together with loan burden variables, were central to the model's decision process.

Two practical findings emerged from the SHAP analysis. First, the model relies heavily on the

external risk score variables, so the quality and fairness of those inputs matter greatly. Second, affordability-related variables such as `credit_term` and other burden measures remain important, showing that the model is not relying on external scores alone.

3.5 Counterfactual Explanations

Counterfactual explanations were generated for three test applicants predicted as defaulters using the DiCE library. These explanations show what minimal changes to an applicant's profile would have flipped the model prediction from default to non-default.

The counterfactual results are useful because they identify changes that are more actionable than a raw model score alone. In particular, changes related to requested credit amount, affordability ratios, and income are more meaningful from a lending perspective than changes involving fixed or sensitive characteristics. The three counterfactual tables from the notebook should be inserted in this section directly, since they provide the clearest side-by-side comparison between each original profile and its counterfactual alternatives.

3.6 Fairness and Responsible Use

A fairness analysis was also conducted on the selected model. Group-wise outcomes were evaluated using a sensitive attribute from the dataset. For each group, the following metrics were reported:

- approval rate
- default rate
- true positive rate
- false positive rate
- precision

Formal fairness summary metrics were also computed.

These results show that profitability and predictive accuracy do not automatically imply fairness. A model can be useful from a business perspective while still distributing approvals or errors unevenly across groups.

A mitigation experiment was also run to examine whether fairness could be improved empirically.

This comparison makes the trade-off between fairness and profitability explicit. A fairer model may reduce profit, while a more profitable model may increase disparities. Responsible model selection therefore requires balancing predictive performance, economic value, and equitable treatment.

4 Limitations and Future Work

4.1 Part 1

Several limitations were present in the Weibull analysis:

- The FD001 subset contained only 100 engines, which limits the stability of parameter

estimation.

- The Weibull model assumes a single dominant failure mechanism, which may oversimplify real engine degradation behaviour.
- Fit quality was assessed mainly visually rather than through formal statistical goodness-of-fit tests.
- Sensor trajectories and operating conditions were not incorporated into the reliability model.

Future work could include formal goodness-of-fit testing, comparison with alternative lifetime distributions, inclusion of covariates from the sensor streams, and larger-scale GPU benchmarking across multiple C-MAPSS subsets.

4.2 Part 2

Several limitations apply to the credit-risk analysis:

- The modelling workflow relied primarily on a single application table and did not use the wider relational structure of the full Home Credit dataset.
- One-hot encoding created a high-dimensional feature space, which can reduce compact interpretability.
- Counterfactual explanations depend on both the fitted model and the DiCE search procedure, so they should be viewed as informative scenarios rather than exact prescriptions.
- Fairness analysis was limited by the sensitive attributes available in the dataset.
- Profit conclusions depend directly on the assumed business cost structure.

Future work could extend the model with richer feature engineering from auxiliary Home Credit tables, compare additional classifiers, calibrate predicted probabilities, test alternative imbalance-handling methods, and explore more advanced fairness-mitigation strategies.

5 Conclusion

This assignment demonstrated two complementary applications of data-driven modelling. In Part 1, Weibull fitting produced interpretable reliability parameters and a practical maintenance threshold, while also illustrating the value of GPU-accelerated optimisation. In Part 2, credit default modelling was evaluated not only in predictive terms, but also through business impact, explainability, counterfactual reasoning, and fairness analysis.

Overall, the assignment showed that useful data science requires more than accurate predictions. It also requires careful preprocessing, economically meaningful evaluation, transparent interpretation, and responsible consideration of how model decisions affect real people.